

Hindi Transliteration Engine — Seq2Seq + Attention

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Abstract

This project converts Hinglish text into Hindi using NLP architectures.

Eight different Seq2Seq and Transformer-based models were trained on 84,000+ text pairs.

The system was evaluated using CER, WER, and BLEU score metrics.

Beam Search decoding improved prediction quality.

The final model was deployed through FastAPI with a custom frontend.

Technologies Used

Technology Stack
TensorFlow, FastAPI, Bahdanau Attention, Transformer, Beam Search

Future Scope

This project can be extended with advanced AI models, cloud deployment, real-time scalability, and improved datasets for better accuracy and performance.

Conclusion

The project demonstrates practical implementation of AI, machine learning, or software technologies to solve real-world problems effectively.